

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

in mind is the concept of handles. The handle acts as the object which passes the values by reference to the various functions. The GUI functions require call-backs and call functions, which direct the pointers to point towards the location of the input as entered or clicked by the user. For example, to execute the program on pressing a button by the user, the code is as given below:

```
function resultpushbutton_  
Callback(hObject, eventdata, handles)
```

The `erroridg(errorstring)` function displays a dialogue box whenever there is error. This function helps in error handling and warning display.

The program flowchart is shown in Fig. 5.

The testing procedure is given below:

1. First launch the MATLAB application from your computer, open `digital_communication.m` and then click on the run button.

2. Once the program is run, GUI program output will open as shown in Fig. 6.

3. The GUI contains four options: FSK, ASK, PSK and Data input.

4. Enter the data in binary form as 0's and 1's. If you enter other data, it will show error message.

This software is very handy for designing parameters like proper sine

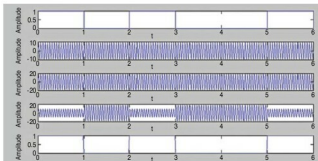


Fig. 8: ASK example output

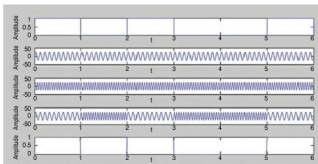


Fig. 9: FSK example output

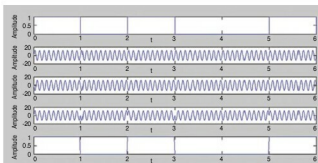


Fig. 10: PSK example output

wave amplitudes and frequencies for the hardware to prevent loading.

The user can enter various combinations of amplitudes and frequencies

to design a noise-free output.

The example program tested by the author with various input parameters is shown in Fig. 7.

The results obtained for ASK, FSK and PSK are shown in Fig. 8, Fig. 9 and Fig. 10, respectively.

One can observe that when ASK is considered, substantial amount of noise is observed, though it is less than with analogue modulation (Fig. 8).

FSK results in less noisy reception of data. Proper designing of frequency and amplitude of the sine waves can result in better reception.

It is observed that PSK has the best reception of data, as phase is least affected by noise. ●

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